

Static Analysis and Verification

Temporal Logics

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SAV 2026, FIT BUT

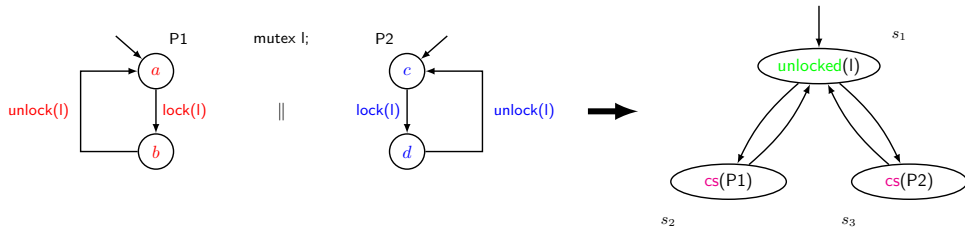
Temporal Logics

CTL*, CTL, LTL

Model of Computation

Kripke Structures

- Informally, **Kripke structures** are directed graphs whose
 - vertices** correspond to configurations of the examined system,
 - the vertices are **labelled** by atomic propositions that are true in the appropriate configurations, and
 - edges** encode possible transitions between the configurations.



- Can be **generated** from the source description of examined systems (or used implicitly as an underlying semantic model of the formulae as well as examined systems).
- The generation involves the **state explosion problem**, or the Kripke structure may be **infinite**—in the following, we, however, concentrate on finite Kripke structures.

Kripke Structures

- Let AP be a set of **atomic propositions** about the configurations of the examined system.
- Formally, a (finite) **Kripke structure** M over AP is a tuple $M = (S, S_0, R, L)$ where
 - ▶ S is a finite set of **states**,
 - ▶ $S_0 \subseteq S$ is a set of **initial states**,
 - ▶ $R \subseteq S \times S$ is a **transition relation**, for convenience supposed to be total (i.e., $\forall s \in S \exists s' \in S: R(s, s')$),
 - ▶ $L: S \rightarrow 2^{AP}$ is a **labelling function** that labels each state by the set of atomic propositions that are true in it.

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 - ▶ $L: S \rightarrow 2^{AP}$ is a **labelling function** that labels each state by the set of atomic propositions that are true in it.
- For the example from the previous slide, we have:
 - ▶ $AP = \{unlocked(l), cs(P1), cs(P2)\}$,
 - ▶ $S = \{s_1, s_2, s_3\}$,
 - ▶ $S_0 = \{s_1\}$,
 - ▶ $R = \{(s_1, s_2), (s_2, s_1), (s_1, s_3), (s_3, s_1)\}$,
 - ▶ $L = \{s_1 \mapsto \{unlocked(l)\}, s_2 \mapsto \{cs(P1)\}, s_3 \mapsto \{cs(P2)\}\}$.

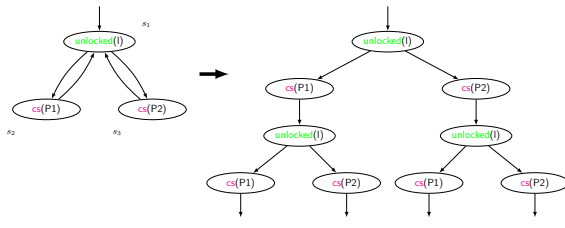
Kripke Structures

- A **path** π in a Kripke structure M is an infinite sequence of states $\pi = s_0 s_1 s_2 \dots$ such that $\forall i \in \mathbb{N}: R(s_i, s_{i+1})$.
- We denote $\Pi(M, s)$ the set of all paths in M that start at $s \in S$.
- The **suffix** π^i of a path $\pi = s_0 s_1 s_2 \dots s_i s_{i+1} s_{i+2} \dots$ is the path $\pi^i = s_i s_{i+1} s_{i+2} \dots$ starting at s_i .

The CTL*Logic

CTL*— Basic Idea

- CTL* formulae describe properties of **computation trees**.
- Infinite computation trees are obtained by **unwinding** a Kripke structure from its initial states.



- CTL* formulae **consist of**:
 - ▶ atomic propositions (= Boolean variables),
 - ▶ Boolean connectives,
 - ▶ **path quantifiers**,
 - ▶ **temporal operators**.

CTL*— Quantifiers and Operators

- **Path quantifiers** — describe the branching structure of a computation tree:
 - ▶ E : for some computation path leading from a state,
 - ▶ A : for all computation paths leading from a state.
- **Temporal operators** — properties of a path through a computation tree:
 - ▶ $X \varphi$ (“next time”, $\bigcirc$): the property φ holds (on the path starting) from the second state of the given path,
 - ▶ $F \varphi$ (“eventually” / “sometimes” (/ “finally”), $\Diamond$): the property φ holds (on the path starting) from some state of the given path,
 - ▶ $G \varphi$ (“always” / “globally”, $\Box$): the property φ holds from all states of the path,
 - ▶ $\varphi U \psi$ (“until”): the property ψ holds from some state of the path, and the property φ holds from all preceding states of the path,
 - ▶ $\varphi R \psi$ (“release”): the property ψ holds from all states of the path up to (and including) the first state from where the property φ holds (if such a state exists).

CTL*— The Syntax

- Let AP be a non-empty set of atomic propositions.
- The syntax of **state formulae**, which are true in a specific state, is given by the following rules:
 - ▶ If $p \in AP$, then p is a state formula.
 - ▶ If φ and ψ are state formulae, then $\neg\varphi$, $\varphi \vee \psi$, $\varphi \wedge \psi$ are state formulae.
 - ▶ If φ is a path formula, then $E\varphi$ and $A\varphi$ are state formulae.
- The syntax of **path formulae**, which are true along a specific path, is given by the following rules:
 - ▶ If φ is a state formula, then φ is a path formula too.
 - ▶ If φ and ψ are path formulae, then $\neg\varphi$, $\varphi \vee \psi$, $\varphi \wedge \psi$, $X\varphi$, $F\varphi$, $G\varphi$, $\varphi U \psi$, and $\varphi R \psi$ are path formulae.
- **CTL*** is the set of **state formulae** generated by the above rules.

CTL*— The Semantics

- Let a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP be given.
- For a *state formula* φ over AP , we denote $M, s \models \varphi$ the fact that φ holds at $s \in S$.
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- Let $s \in S$, π be a path in M , φ_1, φ_2 be state formulae over AP , $p \in AP$, and ψ_1, ψ_2 be path formulae over AP . We **define the relation $\models$ inductively** as follows:

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- For a (state) CTL* formula φ , we write $M \models \varphi$ iff $\forall s_0 \in S_0: M, s_0 \models \varphi$.

CTL*— Basic Operators

- Provided that $AP \neq \emptyset$, it is easy to see that all CTL* operators can be derived from $\vee, \neg, X, U$, and E :
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- Some further connectives may be introduced too, e.g.:
 - ▶ $\varphi \rightarrow \psi \equiv \neg\varphi \vee \psi$,
 - ▶ $\varphi \leftrightarrow \psi \equiv (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$,
 - ▶ ...

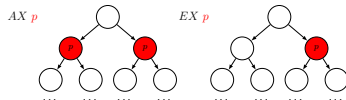
The CTL Logic

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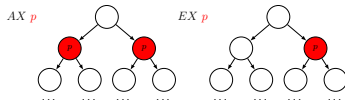
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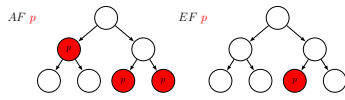


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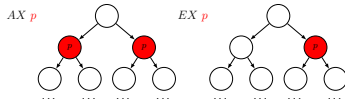
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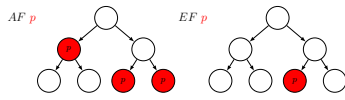
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- CTL is a sublogic of CTL*—the path formulae are restricted to $X \varphi$, $F \varphi$, $G \varphi$, $\varphi U \psi$, and $\varphi R \psi$ for φ, ψ being state formulae.
- In effect, there are allowed these 10 modal CTL operators:

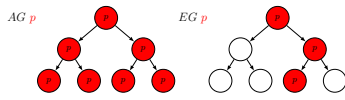
▶ AX and EX ,



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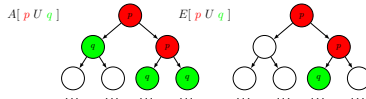
▶ AG and EG ,



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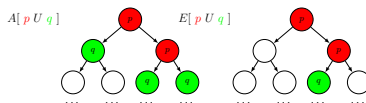
- Continued from the previous slide...
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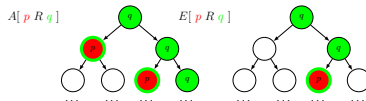
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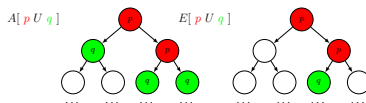
► AR and ER .



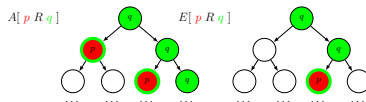
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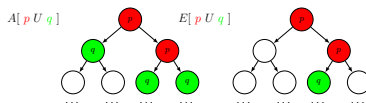
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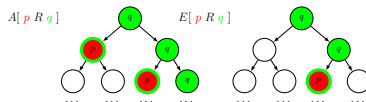
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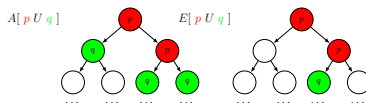
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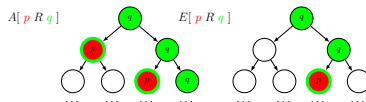
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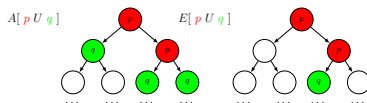
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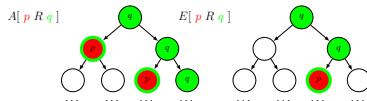
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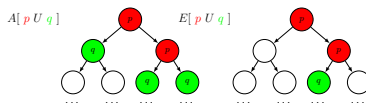
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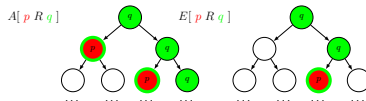
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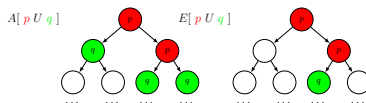
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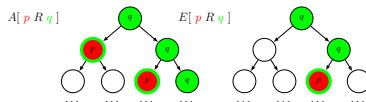
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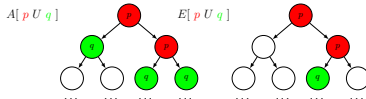
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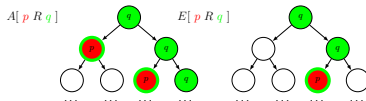
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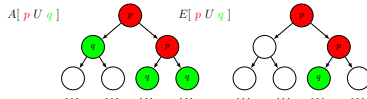
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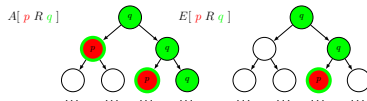
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 - $A[StartAcc R (\neg AccConn)]$

CTL Model Checking

The Basic Idea

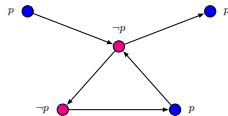
- The **CTL model checking question** to be answered: Given a Kripke structure $M = (S, S_0, R, L)$ over a set of atomic propositions AP and a CTL formula φ over AP , does $M \models \varphi$ hold?
- A **very basic approach** to answer the CTL model checking question by the so-called **explicit-state model checking**:
 - ▶ For every **subformula** ψ of φ , **label by** ψ all those states s of M in which ψ holds (i.e., $M, s \models \psi$).
 - ▶ Perform the labelling **from the inner-most subformulae** (i.e., the most nested ones) going **to the outer ones** exploiting the already computed labels (with atomic propositions corresponding to the original labels of M).
 - ▶ Check whether each state in S_0 gets labelled by φ .
- It is enough to consider the **basic operators** of CTL, i.e., the below syntax for $p \in AP$:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \vee \varphi \mid EX \varphi \mid E[\varphi U \varphi] \mid EG \varphi.$$

Label($\neg\varphi$), Label($\varphi_1 \vee \varphi_2$)

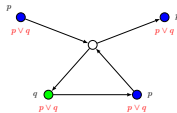
Label($\neg\varphi$)

for all $s \in S$ such that $\varphi \notin \text{Label}(s)$ do
 $\text{Label}(s) := \text{Label}(s) \cup \{\neg\varphi\}$



Label($\varphi_1 \vee \varphi_2$)

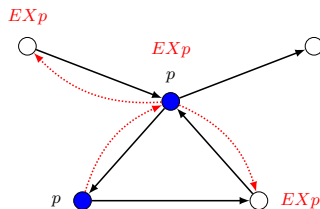
for all $s \in S$ such that $\varphi_1 \in \text{Label}(s)$ or $\varphi_2 \in \text{Label}(s)$ do
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Label($EX\varphi$)

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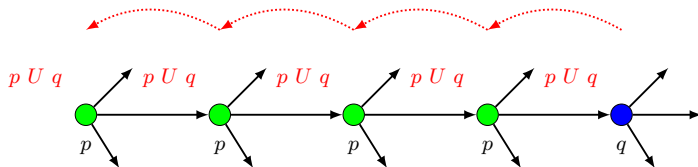
for all $s_2 \in S$ such that $\varphi \in Label(s_2)$ do
 for all $s_1 \in S$ such that $R(s_1, s_2)$ do
 $Label(s_1) := Label(s_1) \cup \{EX\varphi\}$



Label($E[\varphi_1 \text{ U } \varphi_2]$)

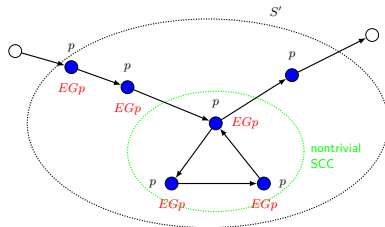
■ The idea:

- ▶ Label first the states already labelled by φ_2 .
- ▶ Look at **predecessors** of states labelled with $\varphi_1 \text{ U } \varphi_2$, and if they are labelled with φ_1 , label them with $\varphi_1 \text{ U } \varphi_2$ as well.



Label($EG\varphi$)

- Based on the following observation: Let $M = (S, S_0, R, L)$ be a Kripke structure, $S' = \{s \in S \mid M, s \models \varphi\}$, and $R' = R \cap (S' \times S')$. For any $s \in S$, $M, s \models EG \varphi$ iff
 - $s \in S'$ and
 - there exists a **path** in the oriented graph $G' = (S', R')$ that leads from s to some node t in a **nontrivial SCC** C of G' .



- An SCC C is **nontrivial** iff either it has more than one node or it contains one node with a self-loop.
- SCCs of a finite oriented graph (V, E) can be computed using **Tarjan's algorithm** in time $O(|E| + |V|)$.

The LTL Logic

LTL — The Syntax

- LTL is another sublogic of CTL* that allows only formulae of the form $A \varphi$ in which the only state subformulae are atomic propositions.
- This is, LTL formulae φ are built according to the grammar:
 - ▶ $\varphi ::= A \psi$ (the use of A is often omitted),
 - ▶ $\psi ::= p \mid \neg\psi \mid \psi \vee \psi \mid \psi \wedge \psi \mid X \psi \mid F \psi \mid G \psi \mid \psi U \psi \mid \psi R \psi$ where $p \in AP$.
- Note that LTL speaks about particular paths in a given Kripke structure only—it ignores its branching structure.
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- Note that LTL speaks about particular paths in a given Kripke structure only—it ignores its branching structure.
- Sometimes, existential LTL allowing formulae of the form $E \varphi$ is used too.
- Note also that while CTL cannot express fairness assumptions (in CTL model checking, they are handled by a special extension of the model checking algorithm), LTL can express fairness assumptions by formulae of the following form:
 - ▶ weak fairness: $(F G \text{ Enabled}) \Rightarrow (G F \text{ Fired})$, i.e., $\Diamond \Box \text{ Enabled} \Rightarrow \Box \Diamond \text{ Fired}$,
 - ▶ strong fairness: $(G F \text{ Enabled}) \Rightarrow (G F \text{ Fired})$, i.e., $\Box \Diamond \text{ Enabled} \Rightarrow \Box \Diamond \text{ Fired}$.

LTL, CTL, and CTL*

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- ▶ CTL cannot express, e.g., the LTL formula $A (FG\ p)$,
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- To complete the picture, here are the complexities of the appropriate model checking algorithms (we will discuss LTL model checking later on):

- ▶ CTL: linear in $|M|$ and linear in $|\varphi|$,
- ▶ LTL and CTL*: linear in $|M|$ and PSPACE-complete in $|\varphi|$,

where $|M| = |S| + |R|$ and $|\varphi|$ is the number of subformulae of φ .

- Finally, as an example of a logic more general than CTL*, we can mention modal μ -calculus based on least/greatest fixpoint operators on sets of states (basically allowing one to define new, specialised modalities).